[INCH-POUND]
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SUPERSEDING
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11 September 1984

COMMERCIAL ITEM DESCRIPTION

ELECTRICAL DISCHARGE MACHINE, TOOL DISINTEGRATING

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-E-80130C for all federal agencies.

1. SCOPE. This commercial item description (CID) covers vibrating head metal disintegrators intended for use in removing broken taps, drills, reamers, bolts, screws or in taking core samples from metals undergoing tests.

2. CLASSIFICATION. The metal disintegrator shall be of the following types. The type to be furnished shall be as specified (see 7.2(b)).

Type I - Drill press mounted
Type II - Portable mounted
Type III - Table mounted

3. SALIENT CHARACTERISTICS

3.1 General requirements. The metal disintegrator shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. Type I metal disintegrators shall consist of a vibrating head having a 0.5-inch arbor for mounting into the chuck of a drill press, a power supply, and a coolant system. Type II metal disintegrators shall consist of a vibrating head, a power supply, a coolant system, and a superstructure. Type III metal disintegrators shall consist of a vibrating head, a power supply, a coolant system, a superstructure, and a worktable.

3.2.1 Vibrating head. The vibrating head shall conduct continuous electric pulses to the electrode that shall cause an electric arc between the electrode and the workpiece, resulting in disintegration of the workpiece. The vibrating head shall hold the electrode by means of collets or a tool holder

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inserted into a chuck. The metal disintegrator shall allow for manual advancement of the electrode into the workpiece. Unless otherwise specified (see 7.2(c)), the quill travel within the vibrating head of the type II and III metal disintegrators shall not be less than 3 inches and 9 inches, respectively. When specified (see 7.2(d)), the metal disintegrator shall have an automatic feed system that advances the electrode into the workpiece by means of either a pneumatic or electrical servomechanism. The automatic feed system shall sense the electrode current and control the feed rate for optimum metal removal.

3.2.2 Power supply. The power supply shall provide control of the voltage and current flowing through the electrode in the vibrating head to the workpiece. Unless otherwise specified (see 7.2(e)), cables connecting the power supply to the vibrating head of type I or II metal disintegrators shall be not less than 8 feet in length. The power supply shall be equipped with a ground fault interrupt circuit breaker.

3.2.3 Coolant system. The coolant system shall filter and pump water through the electrode, remove metal particles, and facilitate the metal disintegration process.

3.2.4 Superstructure. The type II or type III metal disintegrator shall have a superstructure that supports and maintains the position of the vibrating head during operation. The superstructure shall consist of a base, a column, and a crossarm. The superstructure shall permit rotation of the vibrating head in a plane perpendicular to the crossarm and rotation in a plane parallel to the crossarm. The base shall support the column and provide a means of attachment of the superstructure to either a table or other large metal surface. The type II metal disintegrator shall have a base that allows the superstructure to be magnetically attached to a large system from which a broken bolt or tap is to be removed. The type III metal disintegrator shall have either a magnetic base or a base bolted to the superstructure as specified (see 7.2(f)). The force exerted by the magnetic base on a vertical steel plate shall be sufficient to hold the superstructure rigidly in place during operation.

3.2.4.1 Column and crossarm. The column shall be attached to the base and shall support the crossarm. The crossarm shall move along its lengthwise axis, move along the crossarm, and swivel about the column. Movement of the crossarm along its lengthwise axis or along the column shall be by means of a clamp, leadscrew, or rack and pinion as specified (see 7.2(g)). Unless otherwise specified (see 7.2(h)), minimum crossarm axial and column travels shall be as shown in tables I and II for type II and type III metal disintegrators, respectively.

3.2.5 Worktable. The type III metal disintegrator shall be equipped with a worktable. The worktable shall support the workpiece, the superstructure, and the vibrating head. Unless otherwise specified (see 7.2(i)), the dimensions of the worktable shall not be less than 20 inches by 18 inches. Unless otherwise specified (see 7.2(j)), the worktable distributed load capacity shall not be less than 700 pounds.

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TABLE I. Type II metal disintegrator crossarm travel.

Travel mechanism	Axial travel, inches	Column travel, inches
Sliding clamp	16	22
Leadscrew	4	9
Rack and pinion	9	9

TABLE II. Type III metal disintegrator crossarm travel.

Travel mechanism	Axial travel, inches	Column travel, inches
Sliding clamp	16	22
Rack and pinion	9	9

3.2.6 Controls. The metal disintegrator shall include the following controls:

a. Head vibration on/off switch
b. Head vibration intensity control
c. Coolant pump on/off switch
d. Coolant pump pressure control
e. Electrode voltage control
f. Electrode ammeter or other indicating optimum current

3.2.7 Portability. When specified (see 7.2(k)), the type II metal disintegrator power supply and coolant system shall be provided as separate modules with lifting eyes. When specified (see 7.2(l)), type II metal disintegrators shall be equipped with wheels to allow for transport of the power supply and coolant system.

3.2.8 Electrical system. Unless otherwise specified (see 7.2(m)), the electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Machinery, Industrial, Electrical Standard For". Unless otherwise specified, the metal disintegrator shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(n)).

3.2.9 Electric motor. The metal disintegrator’s pump motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturer’s Association (NEMA) MG-1, "Motors and Generators".

3.2.10 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(o)).

3.3 Performance. The tap or bolt removal burn time shall not exceed the performance requirements shown in tables III and IV for the applicable type and performance level. Testing shall be in accordance with sections 5.3 through 5.3.4.

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TABLE III. Type I metal disintegrator performance requirements.

Tap size, inches	Depth, inches	Burn time, minutes
1/4 - 20	1	1-1/2
1/2 - 13	1	4
3/4 - 10	1	12

TABLE IV. Types II and III metal disintegrator performance requirements.

Bolt Size	Depth, inches	Burn time		
		Performance level I, minutes	Performance level II, minutes	Performance level III, minutes
1/2 - 13	1	2	5	10
3/4 - 10	2	5	12	18
1 - 8	2	6	18	30

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(p)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "General Industry, Safety and Health Standards".

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(q)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(r)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(s)), the nameplate shall display the following information:

a. Nomenclature
b. Manufacturer’s name
c. Serial Number
d. Metal disintegrator model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number (if specified)
h. National stock number (if specified)
i. Date of manufacture (if specified)

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3.8 Lubrication plate or chart. If required, a lubrication plate or chart shall be attached to the metal disintegrator when specified (see 7.2(t)). The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The metal disintegrator shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(u)), the metal disintegrator shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test and setup conditions. Unless otherwise specified (see 7.2(v)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(w)).

5.3.2 Operational test. The metal disintegrator shall be operated in accordance with the manufacturer's standard operating procedures. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The metal disintegrator shall be capable of removing taps formed from tool steel with characteristics equivalent to S-4 tool steel with Rockwell hardness of 54C and steel bolts conforming to American Society for Testing and Materials (ASTM) A307 grade A, "Bolts and Studs, Carbon Steel, 60,000 PSI Tensile Strength". The performance tests shall be conducted in accordance with the manufacturer's standard operating procedures. Tap removal shall be

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accomplished using a circular tube electrode to burn through the center portion of the tap, allowing the remaining portions of the tap to collapse into the cavity formed by removing the center. The outer diameter of electrode used to remove 1/4 - 20, 1/2 - 13, and 3/4 - 10 size taps shall be 0.140, 0.250 and 0.406 inches, respectively. Bolt removal shall be accomplished using a circular tube electrode to burn through the bolt, leaving the male threads to be peeled from the female threads. The outer diameter of electrode used to remove 1/2 - 13, 3/4 - 10, and 1 - 8 size bolts shall be 0.375, 0.580 and 0.795 inches, respectively. The threads of the workpiece from which the broken tap or bolt is removed shall be undamaged. The test results will be compared to the requirements corresponding to the selected type and performance level listed in tables III and IV.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(x)).

5.4 Acceptance. Unless otherwise specified (see 7.2(y)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the applicable performance requirements shown in tables III and IV. Failure of the metal disintegrator to meet the requirements shown in tables III and IV for the applicable type and performance level selected shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(z)).

7. NOTES

7.1 Sources of document

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 ASTM standards. Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.6 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

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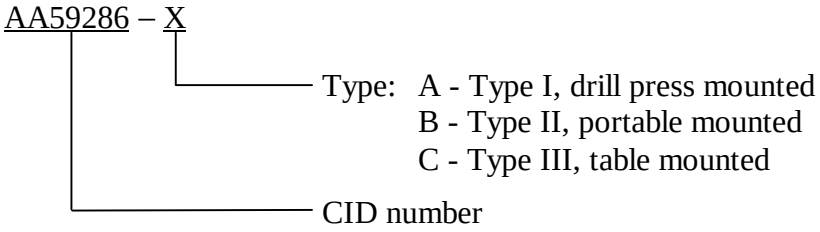
7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Type of metal disintegrator required (see 2)
c. Quill travel, if different (see 3.2.1)
d. Automatic feed system, if required (see 3.2.1)
e. Cable length, if different (see 3.2.2)
f. Superstructure base, as specified (see 3.2.3)
g. Crossarm travel mechanism, as specified (see 3.2.4)
h. Crossarm travel, if different (see 3.2.4)
i. Worktable dimensions, if different (see 3.2.5)
j. Worktable load capacity, if different (see 3.2.5)
k. Modules with lifting eyes, if required (see 3.2.7)
l. Wheels, if required (see 3.2.7)
m. Electrical system industrial standard, if different (see 3.2.8)
n. Electrical system voltage, as specified (see 3.2.8)
o. Optional accessories, as specified (see 3.2.10)
p. Metal disintegrator dimensions and weight, as specified (see 3.4)
q. System of units required (see 3.6)
r. Dual system of units, if required (see 3.6)
s. Nameplate, if different (see 3.7)
t. Lubrication plate or chart, if required (see 3.8)
u. Testing, if different (see 5.3)
v. Test responsibility, if different (see 5.3.1)
w. Certification of calibration, as specified (see 5.3.1)
x. Optional tests, as specified (see 5.3.4)
y. Acceptance, if different (see 5.4)
z. Packaging requirements, if different (see 6.1)

7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - CR4
Navy - MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS
(Project 3410-0062)

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